**PRN – 260245920034**

1. *Display the Name and Title of the Employee who do not have a manager.*

```
mysql> select e.ename as employee_name, e.job as title from emp e where e.mgr is null;
+---------------+-----------+
| employee_name | title     |
+---------------+-----------+
| KING          | PRESIDENT |
+---------------+-----------+
1 row in set (0.00 sec)
```

2. *The sales department is simplifying the pricing policy for all products. The new base price is increasing by the lesser of 0.5 %. Analyze the difference in the two prices.*

```
mysql> select p.descrip as product_name,
    p.price as old_price,
    round(p.price * 1.005, 2) as new_price,
    round((p.price * 1.005) - p.price, 2) as price_difference
from product p
order by p.descrip;
+-------------------------+-----------+-----------+------------------+
| product_name            | old_price | new_price | price_difference |
+-------------------------+-----------+-----------+------------------+
| ACE TENNIS BALLS-3 PACK |       125 |    125.63 |             0.63 |
| ACE TENNIS BALLS-3 PACK |       120 |    120.60 |             0.60 |
| ACE TENNIS BALLS-3 PACK |       500 |    502.50 |             2.50 |
| ACE TENNIS BALLS-6 PACK |       155 |    155.78 |             0.78 |
| ACE TENNIS BALLS-6 PACK |       800 |    804.00 |             4.00 |
| ACE TENNIS BALLS-6 PACK |       100 |    100.50 |             0.50 |
+-------------------------+-----------+-----------+------------------+
18 rows in set (0.04 sec)
```

3. *Display the employee details who have joined on the same day when KING has joined.*

```
mysql> select *
from emp
where hiredate = (
  select hiredate
  from emp
  where ename = 'king'
);
+-------+-------+--------+-----------+------+------------+------+------+--------+---------+-----------+--------+------------+-
---------+
| EMPNO | ENAME | GENDER | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID |
USER NAME | PWD    | PHONE      | isActive |
+-------+-------+--------+-----------+------+------------+------+------+--------+---------+-----------+--------+------------+-
---------+
|  7839 | KING  | M      | PRESIDENT | NULL | 1981-11-17 | 5000 | NULL |     10 |       1 | KING      | r50mpm
| 7132300081 |       1 |
+-------+-------+--------+-----------+------+------------+------+------+--------+---------+-----------+--------+------------+-
---------+
1 row in set (0.00 sec)
```

4. *Find out the difference between maximum salary and salary earned by employee in deptno 10.*

```
mysql> select e.ename as employee_name,
    e.sal as dept10_salary,
    (select max(sal) from emp) - e.sal as salary_difference
from emp e
where e.deptno = 10;
+---------------+---------------+-------------------+
| employee_name | dept10_salary | salary_difference |
+---------------+---------------+-------------------+
| AARAV         |          3350 |              1650 |
| THOMAS        |          1750 |              3250 |
| CLARK         |          2450 |              2550 |
| KING          |          5000 |                 0 |
| MILLER        |          1300 |              3700 |
+---------------+---------------+-------------------+
5 rows in set (0.00 sec)
```

5. *Find out the difference in Rs. Between average earning of dept 20 and 30.*

```
Mysql> select e.ename as employee_name,
        e.sal as salary,
        e.comm as commission
    from emp e
    where e.comm > (0.05 * e.sal);
+---------------+--------+------------+
| employee_name | salary | commission |
+---------------+--------+------------+
| ALLEN         |   1600 |        300 |
| WARD          |   1250 |        500 |
| MARTIN        |   1250 |       1400 |
| GRASS         |   2575 |       2700 |
| MILLER        |   1300 |       1300 |
| AARUSH        |   1350 |       2700 |
| ALEX          |   1250 |        500 |
+---------------+--------+------------+

7 rows in set (0.00 sec)
```

6. *Display name, salary, and comm. For employee whose comm. > 5% of salary.*

```
mysql> select e.ename as employee_name,

        e.sal as salary,  e.comm as commission from emp e

    where e.comm > (0.05 * e.sal);
+---------------+--------+------------+
| employee_name | salary | commission |
+---------------+--------+------------+
| ALLEN         |   1600 |        300 |
```

```
| WARD        |  1250 |      500 |
| MARTIN      |  1250 |     1400 |
| GRASS       |  2575 |     2700 |
| MILLER      |  1300 |     1300 |
| AARUSH      |  1350 |     2700 |
| ALEX        |  1250 |      500 |
+---------------+--------+------------+
7 rows in set (0.00 sec)
```

7. *Find out the employees whose salary is less than the average salary for dept 20.*

```
mysql> select e.ename as employee_name,
              e.sal as salary,
              e.deptno as department_number
       from emp e
       where e.sal < (
         select avg(sal)
         from emp
         where deptno = 20
       ) order by e.deptno, e.sal;

+---------------+--------+-------------------+
| employee_name | salary | department_number |
+---------------+--------+-------------------+
| MILLER        |  1300 |                10 |
| THOMAS        |  1750 |                10 |
| SMITH         |   800 |                20 |
| ADAMS         |  1100 |                20 |
| SHARMIN       |  1750 |                20 |
| BANDISH       |  1750 |                20 |
| JAMES         |   950 |                30 |
| WARD          |  1250 |                30 |
+---------------+--------+-------------------+
20 rows in set (0.00 sec)
```

8. *Display information of employee who has no people reporting to them.*

```
mysql> select e.ename as employee_name,
              e.job as title,
              e.sal as salary,
              e.comm as commission,
              e.deptno as department_number
       from emp e
       where e.empno not in (
         select distinct mgr
         from emp where mgr is not null) order by e.deptno, e.ename;
+---------------+---------------------+--------+------------+-------------------+
| employee_name | title               | salary | commission | department_number |
+---------------+---------------------+--------+------------+-------------------+
| AARAV         | CLERK               |  3350 |       NULL |                10 |
```

```
| MILLER      | CLERK            |  1300 |     1300 |            10 |
| THOMAS      | CLERK            |  1750 |     NULL |            10 |
| ADAMS       | CLERK            |  1100 |     NULL |            20 |
| BANDISH     | Public Relation  |  1750 |     NULL |            20 |
| FRED        | vice PRESIDENT   |  5000 |     NULL |            20 |
| SHARMIN     | Public Relation  |  1750 |     NULL |            20 |
| SMITH       | CLERK            |   800 |     NULL |            20 |
+---------------+--------------------+--------+-----------+-------------------+
21 rows in set (0.00 sec)
```

9. *Display information of employees who is having maximum number of people reporting to them.*

```
mysql> select e.ename as manager_name,
                e.job as title,
                e.sal as salary,
                e.deptno as department_number,
                count(sub.empno) as number_of_reports
            from emp e
            join emp sub
            on e.empno = sub.mgr
            group by e.empno, e.ename, e.job, e.sal, e.deptno
            having count(sub.empno) = (
              select max(report_count)
              from (
                select count(*) as report_count
                from emp
                where mgr is not null
                group by mgr
              ) t
            );
+--------------+----------+--------+------------------+-------------------+
| manager_name | title    | salary | department_number | number_of_reports |
+--------------+----------+--------+------------------+-------------------+
| GRASS        | SALESMAN |  2575 |             30 |                 8 |
+--------------+----------+--------+------------------+-------------------+
1 row in set (0.06 sec)
```

10. *Give a query to concatenate empno, ename and mgr with spaces between.*

```
mysql> select concat_ws(' ', empno, ename, mgr) as employee_info from emp;
+--------------------+
| employee_info      |
+--------------------+
| 6001 GITA 7920     |
| 6129 VRUSHALI 7920 |
| 6473 SHARMIN 7920  |
| 6781 BANDISH 7920  |
| 7369 SMITH 7902    |
| 7415 AARAV 7902    |
```

```
+--------------------------+
30 rows in set (0.01 sec)
```

11. *List the employee who is hired the earliest and least.*

```
mysql> select * from emp where hiredate = (select min(hiredate) from emp);
+-------+--------+--------+----------+------+------------+------+------+--------+---------+-----------+------------+-------+--------
---+
| EMPNO | ENAME  | GENDER | JOB      | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER
NAME | PWD      | PHONE | isActive |
+-------+--------+--------+----------+------+------------+------+------+--------+---------+-----------+------------+-------+--------
---+
| 7920 | GRASS  | M     | SALESMAN | 7919 | 1980-02-14 | 2575 | 2700 |    30 |      5 | GRASS    | sales@2017
| NULL |      1 |
| 7945 | AARUSH | M     | SALESMAN | 7902 | 1980-02-14 | 1350 | 2700 |    30 |    NULL | AARUSH    |
ABCDEF    | NULL |      0 |
+-------+--------+--------+----------+------+------------+------+------+--------+---------+-----------+------------+-------+--------
---+
2 rows in set (0.00 sec)
```

12. *What length of longest ename and by how many characters is it long than its nearest one?*

```
mysql> select max(length(ename)) as longest_length,
                max(length(ename)) - (
                  select max(length(ename))
                  from emp
                  where length(ename) < (
                    select max(length(ename)) from emp
                  )
                ) as difference_from_nearest
            from emp;
+----------------+-------------------------+
| longest_length | difference_from_nearest |
+----------------+-------------------------+
|              8 |                       1 |
+----------------+-------------------------+
1 row in set (0.00 sec)
```

13. *Which employees earn less than 30% of president s salary?*

```
mysql> select e.ename as employee_name,
                e.sal as salary,
                e.deptno as department_number
            from emp e
            where e.sal < (
              select 0.3 * sal
              from emp
              where job = 'president'
            )order by e.sal;
```

```
+---------------+--------+-------------------+
| employee_name | salary | department_number |
+---------------+--------+-------------------+
| SMITH         |   800 |                20 |
| JAMES         |   950 |                30 |
| Neel          |  1000 |                60 |
| ADAMS         |  1100 |                20 |
| WARD          |  1250 |                30 |
| MARTIN        |  1250 |                30 |
| ALEX          |  1250 |                30 |
| MILLER        |  1300 |                10 |
| AARUSH        |  1350 |                30 |
+---------------+--------+-------------------+
9 rows in set (0.00 sec)
```

14. *List the department spending highest on salary.*

```
mysql > select deptno, sum(sal) as total_salary
    from emp group by deptno
    having sum(sal) = (
      select max(total) from (select sum(sal) as total from emp group by deptno) t
    );
+--------+--------------+
| deptno | total_salary |
+--------+--------------+
|    30 |        22225 |
+--------+--------------+
1 row in set (0.00 sec)
```

15. *List the department having least number of staff.*

```
mysql> select deptno,
                 count(*) as staff_count
           from emp
           group by deptno
           having count(*) = (
             select min(staff_count)
             from (
                select count(*) as staff_count
                from emp
                group by deptno
             ) t
           );
+--------+-------------+
| deptno | staff_count |
+--------+-------------+
|    60 |           2 |
+--------+-------------+
1 row in set (0.00 sec)
```

16. *List the department having neither the highest not the lowest of avg salary.*

```
mysql> select deptno, avg(sal) as avg_salary from emp
            group by deptno order by avg(sal) limit 1 offset 1;
+--------+------------+
| deptno | avg_salary |
+--------+------------+
|    50 |  1750.0000 |
+--------+------------+
1 row in set (0.05 sec)
```

17. *Display details of all employees as Mr. Ford works under <manger name>.*

```
Mysql> select concat('mr. ', e.ename, ' works under ', m.ename) as employee_info
            from emp e
            join emp m
            on e.mgr = m.empno
            where e.ename = 'ford';
+----------------------------+
| employee_info              |
+----------------------------+
| mr. FORD works under JONES |
+----------------------------+
1 row in set (0.03 sec)
```

18. *Display employee information who have joined the organization before 'BLAKE'*

```
mysql> select * from emp where hiredate < (select hiredate  from emp where ename = 'blake');
+-------+--------+--------+----------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
| EMPNO | ENAME  | GENDER | JOB      | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER NAME    | PWD        | PHONE      |
isActive |
+-------+--------+--------+----------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
|  7369 | SMITH  | M      | CLERK    | 7902 | 1980-12-17 |  800 | NULL |     20 |       2 | TRITONE      | a12recmpm  | 7032300023 |        0 |
|  7499 | ALLEN  | M      | SALESMAN | 7698 | 1981-02-20 | 1600 |  300 |     30 |       4 | ALWAYS TESTE | sales@2017 | 7032300096 |        1 |
|  7521 | WARD   | M      | SALESMAN | 7698 | 1981-02-22 | 1250 |  500 |     30 |       1 | WARD         | sales@2017 | 7132300034 |        1 |
|  7566 | JONES  | M      | MANAGER  | 7839 | 1981-04-02 | 2975 | NULL |     20 |       4 | HONEYCOMB    | a12recmpm  | 7132300039 |        1 |
|  7920 | GRASS  | M      | SALESMAN | 7919 | 1980-02-14 | 2575 | 2700 |     30 |       5 | GRASS        | sales@2017 | NULL       |        1 |
|  7945 | AARUSH | M      | SALESMAN | 7902 | 1980-02-14 | 1350 | 2700 |     30 |    NULL | AARUSH       | ABCDEF     | NULL       |        0 |
+-------+--------+--------+----------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
6 rows in set (0.00 sec)
```

19. *Display employee information that has joined the organization before their subordinates.*

```
mysql> select distinct m.empno, m.ename, m.job, m.hiredate, m.sal, m.deptno
            from emp m join emp e on m.empno = e.mgr
            where m.hiredate < e.hiredate;
+-------+-------+-----------+------------+------+--------+
| empno | ename | job       | hiredate   | sal  | deptno |
+-------+-------+-----------+------------+------+--------+
|  7920 | GRASS | SALESMAN  | 1980-02-14 | 2575 |     30 |
|  7902 | FORD  | ANALYST   | 1981-12-03 | 3000 |     20 |
|  7698 | BLAKE | MANAGER   | 1981-05-01 | 2850 |     30 |
+-------+-------+-----------+------------+------+--------+
7 rows in set (0.00 sec)
```

20. *List the employee (s) who earns the highest.*

```
mysql> select *
from emp
where sal = (
   select max(sal)
   from emp
);
```

```
+-------+-------+--------+----------------+------+------------+------+------+--------+---------+------------+-----------+---------+
| EMPNO | ENAME | GENDER | JOB            | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER
NAME  | PWD      | PHONE      | isActive |
+-------+-------+--------+----------------+------+------------+------+------+--------+---------+------------+---------+----------+
|  7839 | KING  | M      | PRESIDENT      | NULL | 1981-11-17 | 5000 | NULL |     10 |       1 | KING       | r50mpm
| 7132300081 |       1 |
|  7840 | FRED  | M      | vice PRESIDENT | 7839 | 1983-08-09 | 5000 | NULL |     20 |       1 | SAND STONE |
a12recmpm | NULL      |       1 |
+-------+-------+--------+----------------+-----+---------+------------+-----------+------------+----------+
2 rows in set (0.00 sec)
```

21. *List the mgr in dept 10 along with their dept details as well manager refers to anyone to whom at least 1 person reports.*

```
mysql> select distinct m.empno,
     m.ename as manager_name,
     m.job,
     m.deptno,
     d.dname as department_name,
     d.loc as department_location
from emp m
join emp e
  on m.empno = e.mgr
join dept d
  on m.deptno = d.deptno
where m.deptno = 10;
+-------+--------------+-----------+--------+----------------+---------------------+
| empno | manager_name | job       | deptno | department_name | department_location |
+-------+--------------+-----------+--------+----------------+---------------------+
|  7782 | CLARK        | MANAGER   |     10 | ACCOUNTING     | NEW YORK            |
|  7839 | KING         | PRESIDENT |     10 | ACCOUNTING     | NEW YORK            |
+-------+--------------+-----------+--------+----------------+---------------------+
2 rows in set (0.00 sec)
```

22. *In which year did the most people join the organization? Display the year and the number of employees.*

```
mysql> select year(hiredate) as join_year,
     count(*) as num_employees
from emp
group by year(hiredate)
order by num_employees desc
limit 1;
```

```
+-----------+---------------+
| join_year | num_employees |
+-----------+---------------+
|     1981 |          19 |
+-----------+---------------+
1 row in set (0.00 sec)
```

23. *List number, name, job, manager number and manager's job of each employee reporting to them*

```
select e.empno,
    e.ename,
    e.job,
    e.mgr as manager_number,
    m.job as manager_job
from emp e
left join emp m
  on e.mgr = m.empno;
+-------+----------+--------------------+----------------+-------------+
| empno | ename    | job                | manager_number | manager_job |
+-------+----------+--------------------+----------------+-------------+
|  6001 | GITA     | Compliance officers |          7920 | SALESMAN    |
|  6129 | VRUSHALI | Compliance officers |          7920 | SALESMAN    |
|  6473 | SHARMIN  | Public Relation     |          7920 | SALESMAN    |
|  6781 | BANDISH  | Public Relation     |          7920 | SALESMAN    |
|  7369 | SMITH    | CLERK               |          7902 | ANALYST     |
|  7415 | AARAV    | CLERK               |          7902 | ANALYST     |
|  7421 | THOMAS   | CLERK               |          7920 | SALESMAN    |
+-------+----------+--------------------+----------------+-------------+
30 rows in set (0.00 sec)
```

24. *Display information about employees who have the maximum number of employees reporting to him*

```
mysql> select m.empno,
    m.ename as manager_name,
    m.job as manager_job,
    count(e.empno) as num_subordinates
from emp m
join emp e on m.empno = e.mgr
group by m.empno, m.ename, m.job
order by num_subordinates desc
limit 1;
+-------+--------------+-------------+------------------+
| empno | manager_name | manager_job | num_subordinates |
+-------+--------------+-------------+------------------+
|  7920 | GRASS        | SALESMAN    |                8 |
+-------+--------------+-------------+------------------+
1 row in set (0.00 sec)
```

25. *Display the details of all employees who report to BLAKE*

```
Mysql> select e.* from emp e join emp m on e.mgr = m.empno where m.ename = 'blake';
+-------+--------+--------+----------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
| EMPNO | ENAME  | GENDER | JOB      | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER NAME    | PWD        |
PHONE     | isActive |
+-------+--------+--------+----------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
|  7499 | ALLEN  | M      | SALESMAN | 7698 | 1981-02-20 | 1600 |  300 |     30 |       4 | ALWAYS TESTE | sales@2017 |
7032300096 |      1 |
| 7521 | WARD   | M     | SALESMAN | 7698 | 1981-02-22 | 1250 |  500 |    30 |      1 | WARD       | sales@2017 | 7132300034 |
1 |
|  7654 | MARTIN | M      | SALESMAN | 7698 | 1981-09-28 | 1250 | 1400 |     30 |       6 | LIFE RACER   | sales@2017 |
7132300050 |      1 |
| 7844 | TURNER | M     | SALESMAN | 7698 | 1981-09-08 | 1500 |    0 |    30 |      5 | SAND DUST    | sales@2017 | NULL       |
1 |
|  7900 | JAMES  | M      | CLERK    | 7698 | 1981-12-03 |  950 | NULL |     30 |       2 | CRAZY LEADER | sales@2017 |
7132300042 |      1 |
|  7949 | ALEX   | M      | MANAGER  | 7698 | 1982-01-24 | 1250 |  500 |     30 |       5 | QUARTERREST  | sales@2017 |
7156567183 |      1 |
+-------+--------+--------+----------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
6 rows in set (0.04 sec)
```

26. *Display the department no of all departments having more than 1 clerks*

```
mysql> select deptno
from emp
where job = 'clerk'
group by deptno
having count(*) > 1;
+--------+
| deptno |
+--------+
|    10 |
|    20 |
|    30 |
+--------+
3 rows in set (0.00 sec)
```

27. *Which employees earn less than 30% of the president's salary?*

```
mysql> select *
from emp
where sal < (
   select sal * 0.3
   from emp
   where job = 'president'
);
+-------+--------+--------+----------------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
| EMPNO | ENAME  | GENDER | JOB            | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER NAME    | PWD        |
| PHONE     | isActive |
+-------+--------+--------+----------------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
|  7369 | SMITH  | M      | CLERK          | 7902 | 1980-12-17 |  800 | NULL |     20 |       2 | TRITONE      | a12recmpm  |
7032300023 |      0 |
|  7521 | WARD   | M      | SALESMAN       | 7698 | 1981-02-22 | 1250 |  500 |     30 |       1 | WARD         | sales@2017 |
7132300034 |      1 |
```

```
| 7654 | MARTIN | M      | SALESMAN       | 7698 | 1981-09-28 | 1250 | 1400 |     30 |       6 | LIFE RACER   | sales@2017 |
7132300050 |      1 |
| 7876 | ADAMS  | M      | CLERK          | 7788 | 1983-01-12 | 1100 | NULL |     20 |       1 | CRAZY ANYONE | a12recmpm  |
7132300086 |      1 |
| 7889 | Neel   | M      | vice PRESIDENT | 7839 | 1983-08-09 | 1000 | NULL |     60 |       0 | Neel         | BRD@1122   | NULL       |
1 |
| 7900 | JAMES  | M      | CLERK          | 7698 | 1981-12-03 |  950 | NULL |     30 |       2 | CRAZY LEADER | sales@2017 |
7132300042 |      1 |
| 7934 | MILLER | M      | CLERK          | 7782 | 1982-01-23 | 1300 | 1300 |     10 |       2 | QUARTERNOTE  | r50mpm     |
7132300055 |      0 |
| 7945 | AARUSH | M      | SALESMAN       | 7902 | 1980-02-14 | 1350 | 2700 |     30 |    NULL | AARUSH       | ABCDEF     | NULL
|      0 |
| 7949 | ALEX   | M      | MANAGER        | 7698 | 1982-01-24 | 1250 |  500 |     30 |       5 | QUARTERREST  | sales@2017 |
7156567183 |      1 |
+-------+--------+--------+----------------+------+------------+------+------+--------+---------+--------------+------------+------------+----------+
9 rows in set (0.00 sec)
```

28. Display all employee names and the locations of their departments except 'CHICAGO'

```
mysql> select e.ename,
       d.loc as department_location
from emp e
join dept d on e.deptno = d.deptno
where d.loc <> 'chicago';
+----------+---------------------+
| ename    | department_location |
+----------+---------------------+
| AARAV    | NEW YORK            |
| THOMAS   | NEW YORK            |
| CLARK    | NEW YORK            |
| KING     | NEW YORK            |
| MILLER   | NEW YORK            |
| SHARMIN  | DALLAS              |
| BANDISH  | DALLAS              |
| SMITH    | DALLAS              |
+----------+---------------------+
18 rows in set (0.00 sec)
```

29. Display department that has no employees

```
mysql> select d.deptno, d.dname, d.loc from dept d
where d.deptno not in (
   select distinct deptno
   from emp
);
+--------+------------+--------+
| deptno | dname      | loc    |
+--------+------------+--------+
|     40 | OPERATIONS | BOSTON |
+--------+------------+--------+
1 row in set (0.03 sec)
```

30. *Display the products purchased by JOCK SPORTS which are not purchased by JUST TENNIS*

```
mysql> select p.prodid, p.descrip, p.price from product p join sales s on p.prodid = s.prodid
            where s.custname = 'jock sports' and p.prodid not in (
            select prodid  from sales where custname = 'just tennis' );
Empty set (0.01 sec)
```

31. *Write a query to find out the employees who are first to join their departments*

```
mysql> select e.empno,
     e.ename,
     e.job,
     e.hiredate,
     e.deptno
from emp e
where e.hiredate = (
   select min(hiredate)
   from emp
   where deptno = e.deptno
);
+-------+----------+---------------------+------------+--------+
| empno | ename    | job                 | hiredate   | deptno |
+-------+----------+---------------------+------------+--------+
|  6129 | VRUSHALI | Compliance officers | 1981-07-19 |     50 |
|  7369 | SMITH    | CLERK               | 1980-12-17 |     20 |
|  7782 | CLARK    | MANAGER             | 1981-06-09 |     10 |
|  7888 | Ruhan    | Network             | 1983-08-09 |     60 |
+-------+----------+---------------------+------------+--------+
9 rows in set (0.00 sec)
```

32. *Create a query to display the name, hiredate, and salary for all employees who have both the same salary and commission as Scott.*

```
mysql> select ename,
     hiredate,
     sal,
     comm
from emp
where sal = (select sal from emp where ename = 'SCOTT')
  and comm = (select comm from emp where ename = 'SCOTT');
Empty set (0.00 sec)
```

33. *List all the employees who have at least one person reporting to them.*

```
mysql> select distinct m.empno,
     m.ename,
     m.job,
     m.deptno
from emp m
where m.empno in (
```

```
   select mgr
   from emp
   where mgr is not null
);

+-------+---------+-----------+--------+
| empno | ename   | job       | deptno |
+-------+---------+-----------+--------+
|  7566 | JONES   | MANAGER   |     20 |
|  7698 | BLAKE   | MANAGER   |     30 |
|  7782 | CLARK   | MANAGER   |     10 |
|  7788 | SCOTT   | ANALYST   |     20 |
|  7839 | KING    | PRESIDENT |     10 |
+-------+---------+-----------+--------+
9 rows in set (0.00 sec)
```

34. *List the name of the employees with their immediate higher authority.*

```
mysql> select e.ename as employee_name,
     m.ename as manager_name
from emp e
left join emp m
  on e.mgr = m.empno;
+---------------+--------------+
| employee_name | manager_name |
+---------------+--------------+
| GITA          | GRASS        |
| VRUSHALI      | GRASS        |
| SHARMIN       | GRASS        |
| BANDISH       | GRASS        |
| SMITH         | FORD         |
| AARAV         | FORD         |
+---------------+--------------+
30 rows in set (0.00 sec)
```

35. *List all the employees who do not manage any one.*

```
mysql> select e.empno,
     e.ename,
     e.job,
     e.deptno
from emp e
where e.empno not in (
   select distinct mgr
   from emp
   where mgr is not null
);

+-------+----------+---------------------+--------+
| empno | ename    | job                 | deptno |
+-------+----------+---------------------+--------+
```

```
| 6001 | GITA     | Compliance officers |     30 |
| 6129 | VRUSHALI | Compliance officers |     50 |
| 6473 | SHARMIN  | Public Relation     |     20 |
| 6781 | BANDISH  | Public Relation     |     20 |
| 7369 | SMITH    | CLERK               |     20 |
| 7415 | AARAV    | CLERK               |     10 |
| 7421 | THOMAS   | CLERK               |     10 |
+-------+----------+---------------------+--------+
21 rows in set (0.00 sec)
```

36. *Write a query to display a '*' against the row of the most recently hired employee.*

```
mysql> select e.empno,
     e.ename,
     e.job,
     e.hiredate,
     e.sal,
     case
       when e.hiredate = (select max(hiredate) from emp)
       then '*'
       else ''
     end as marker
from emp e;

+-------+----------+---------------------+------------+------+--------+
| empno | ename    | job                 | hiredate   | sal  | marker |
+-------+----------+---------------------+------------+------+--------+
| 6001 | GITA     | Compliance officers | 1981-07-19 | 1750 |        |
| 6129 | VRUSHALI | Compliance officers | 1981-07-19 | 1750 |        |
| 6473 | SHARMIN  | Public Relation     | 1981-07-19 | 1750 |        |
| 6781 | BANDISH  | Public Relation     | 1981-07-19 | 1750 |        |
| 7369 | SMITH    | CLERK               | 1980-12-17 | 800  |        |
| 7415 | AARAV    | CLERK               | 1981-12-31 | 3350 |        |
+-------+----------+---------------------+------------+------+--------+
30 rows in set (0.00 sec)
```

37. *Write a correlated sub-query to list out the employees who earn more than the average salary of their department.*

```
select e.empno,
     e.ename,
     e.job,
     e.sal,
     e.deptno
from emp e
where e.sal > (
   select avg(sal)
   from emp
   where deptno = e.deptno
);
```

```
+-------+---------+----------------+------+--------+
| empno | ename   | job            | sal  | deptno |
+-------+---------+----------------+------+--------+
|  7415 | AARAV   | CLERK          | 3350 |     10 |
|  7566 | JONES   | MANAGER        | 2975 |     20 |
|  7698 | BLAKE   | MANAGER        | 2850 |     30 |
|  7788 | SCOTT   | ANALYST        | 3000 |     20 |
|  7839 | KING    | PRESIDENT      | 5000 |     10 |
|  7840 | FRED    | vice PRESIDENT | 5000 |     20 |
|  7888 | Ruhan   | Network        | 2000 |     60 |
|  7902 | FORD    | ANALYST        | 3000 |     20 |
|  7919 | HOFFMAN | MANAGER        | 4150 |     30 |
|  7920 | GRASS   | SALESMAN       | 2575 |     30 |
+-------+---------+----------------+------+--------+
10 rows in set (0.00 sec)
```

38. *Find the 2nd maximum salary.*

```
mysql> select distinct sal
from emp
order by sal desc
limit 1 offset 1;
+------+
| sal  |
+------+
| 4150 |
+------+
1 row in set (0.00 sec)
```

39. *Select the duplicate records (Records, which are inserted, that already exist) in the EMP table.*

```
mysql> select empno, ename, job, mgr, hiredate, sal, comm,  deptno,
    count(*) as duplicate_count
from emp
group by empno, ename, job, mgr, hiredate, sal, comm, deptno
having count(*) > 1;
Empty set (0.00 sec)
```

40. *List the Employees in Deptno 10 other than King.*

```
mysql> select empno,
    ename,
    job,
    hiredate,
    sal
from emp
where deptno = 10
 and ename <> 'KING';
```

```
+-------+--------+---------+------------+------+
| empno | ename  | job     | hiredate   | sal  |
+-------+--------+---------+------------+------+
|  7415 | AARAV  | CLERK   | 1981-12-31 | 3350 |
|  7421 | THOMAS | CLERK   | 1981-07-19 | 1750 |
|  7782 | CLARK  | MANAGER | 1981-06-09 | 2450 |
|  7934 | MILLER | CLERK   | 1982-01-23 | 1300 |
+-------+--------+---------+------------+------+
4 rows in set (0.00 sec)
```